

Celestial sphere c. 1700

This armillary sphere is an astronomical teaching device, developed by Greek and Arab astronomers. It is a skeletal representation of the “celestial sphere” used for measuring astronomical coordinates. From the early 17th century onward, the Sun was placed at the center of the sphere, as shown here.

Ring measures declination, the equivalent of latitude

Plane of the solar system

equivalents of today's natural history museums.

Exotic plant species brought back from around the world were cultivated in botanical gardens, and the collection of new specimens became an increasingly important part of exploratory and trading missions overseas.

The growing interest in novelty undermined the reliance on the ancients, as it became clear that they had not, after all, known everything. The classical geographers had nothing to say about the new lands now being discovered, and since the authorities were also silent about the many new species of plant and animal, the scholars of the time had no choice but to investigate for themselves, and start to draw their own conclusions.

Written in the stars

Improved technology was also undermining other long-held theories. The most celebrated casualty of the entire revolution was the ancient, Earth-centered model of the universe.

Italian physicist and astronomer Galileo Galilei is often credited with discovering that Earth and the other planets orbit the Sun, using one of the first telescopes. However, the idea had been in the air for several decades (see left). Galileo's discovery in 1610 of the four moons orbiting

Jupiter showed that Earth was not the center of all motion in the universe, and his observation of the Moon-like phases of Venus showed that it must be orbiting the Sun. The most persuasive evidence for this theory came from the observations of Danish



Timekeepers

Improvements in time measurement aided both astronomy and navigation. The weight-driven brass lantern clock was one step forward, although this Japanese version from the mid-17th century still does not have a minute hand. The pendulum clock invented by Christiaan Huygens in 1656 marked a further advance in precision.

astronomer Tycho Brahe. He used another refined form of technology—a precise measuring device known as a mural quadrant, developed from Arabic astrolabes (see p.175).

However, these observations alone could not have created a new theory—they merely disproved the old one. The crucial leap was made by Brahe's one-time student, German astronomer Johannes Kepler. Careful study of Brahe's measurements allowed him to develop a set of laws, published in 1609, that showed how the planets follow elliptical, rather than circular, orbits around the Sun. The acceptance of Kepler's laws involved a brutal divorce from classical ideas of circular perfection, and was not helped by the lack of a model to explain what force kept the planets on their elliptical courses—that had to wait until the century's end, and the breakthroughs of Isaac Newton (see above).

Advances in biology

While astronomy and physics were undergoing their revolution, medicine and anatomy were also in a period of rapid change. The discovery of a lost text by Galen in the early



Life through a lens

The first book of microscope observations was Robert Hooke's *Micrographia* of 1664, which described and illustrated Hooke's observations of insects, plants, and living cells. This illustration of an ant shows more detailed anatomy than was previously possible.

ENGLISH PHYSICIST (1643–1727)

ISAAC NEWTON

Kepler's laws explained how planets and other celestial objects followed their orbits, but could not explain why. The realization that the force of gravity found on Earth might extend out into space, and be generated by other objects of sufficient mass, came from Isaac Newton.

Newton was a brilliant but tempestuous individual, fascinated by mathematics, astronomy, optics, and alchemy. During the last two decades of his life, he was the most revered “natural philosopher” in Europe. The universal laws of motion and gravitation, and the model of the behavior of light that he discovered, remained the foundations of physics for two centuries.



NEWTON'S TELESCOPE

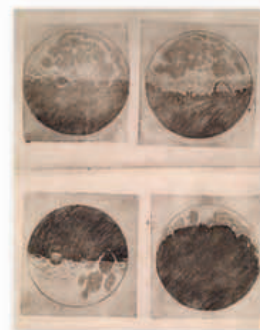
16th century intrigued the Flemish anatomist Andreas Vesalius, who discovered that Galen had clearly never dissected a human body. Vesalius' great atlas of anatomy, *De Humani Corporis Fabrica* (1543), inspired a new generation to continue the study of anatomy without undue deference to authorities from the past.

One great battleground between traditionalism and the new approach was the nature of the circulatory system. Galen taught that the blood did not circulate through the body, but rose and fell in “tides,” along with the body's other “humors” (phlegm, and black and yellow bile). Imbalances in the humors supposedly caused a variety of ailments, and much of medieval medicine focused on how best to regulate them.

Only when Galen's description of the heart was proved wrong did physicians begin to question this theory. In 1603, Italian anatomist Hieronymus Fabricius discovered valves within certain blood vessels, but it was not until 1628 that William Harvey, personal physician to Charles I of England, announced his discovery that the valves only worked in one direction. Blood vessels must therefore form two distinct groups—arteries taking blood away from the heart, and veins carrying blood back toward it.

Harvey's discovery met with fierce resistance, not only because it undermined Galen, but also because, with no visible connection between

arteries and veins, it relied on tiny “capillaries,” too small to be seen, to complete the system. By stating the evidence rather than attacking rival ideas, Harvey formed a template for later scientific publications. In 1661, improvements to the recently invented microscope finally allowed direct observation of the capillaries, vindicating Harvey and confirming the supremacy of the new method of observation, experiment, and deduction.



Mountains on the Moon

Using a telescope Galileo discovered the varying terrain of the Moon (shown here), and spots on the Sun.

AFTER

After the scientific revolution, science became established as the best method to explain and predict natural phenomena.

SCIENCE AND SOCIETY

The 17th century saw the foundation of many scientific societies across Europe. Establishment of true **scientific method** inspired thinkers in many other fields, giving rise to economic, social, and political sciences among others.

THE RATIONAL UNIVERSE

Newton's “clockwork” model of the universe following immutable physical laws inspired Enlightenment philosophers **270–71** >>, yet left many in fear of an essentially meaningless Universe. **Darwin's theory of evolution 340–41** >> had a similar traumatic effect.

TECHNOLOGY AND EMPIRE

The rise of science led to rapid technological advance in Western Europe, culminating in the **industrial revolution 292–93** >>. The growing supremacy of Western technology ensured longevity of the nascent European empires.